Ethan Collins

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EDUCATION

The University of Texas at Dallas Richardson, TX
Bachelor of Science, Software Engineering *Expected May 2028*

PROFESSIONAL EXPERIENCE

Auraion.ai — Lead UX/Full Stack Intern — *Paid Internship* June 2026 – August 2026

- Architected a student-facing dual-interface Progressive Web App and internal operations dashboard using **Next.js (App Router)**, **TypeScript**, and **Tailwind CSS**, tuning UI density and SSR component specs separately for each audience.
- Engineered a COPPA/FERPA-compliant onboarding pipeline backed by **Supabase (PostgreSQL, Auth, RLS)** with conversational intake and automated parent-consent routing, cutting account activation to under three minutes.
- Developed an event-driven adaptive UI engine in **TypeScript** that ingests real-time student engagement telemetry to dynamically trigger contextual hints, switch input modalities, and reset workspace state automatically.

TECHNICAL PROJECTS

LowBall (FastAPI, React Native, TypeScript, Gemini API, aiosqlite) — *AI Marketplace Negotiation Companion* May 2026

- Architected a 100% async **FastAPI** and **aiosqlite** backend with a unified "Council-of-Thought" multi-persona agent and real-time **WebSocket** streaming, cutting multi-turn negotiation inference latency by ∼65%.
- Engineered a multimodal vision scraper, token-bucket rate limiter, and resilience layer with an automated **Pydantic-to-TypeScript** schema generator to enforce end-to-end type safety across mobile and backend services.

EarlyAxxess (React.js, Neo4j, Gemini API, PrimeKG, Hetionet) — *GraphRAG Medical Consultation & EMR System* February 2026

- Built a clinical **GraphRAG** pipeline integrating **Neo4j** with biomedical knowledge graphs (**PrimeKG** and **Hetionet**) to ground **Gemini API** inference and eliminate hallucinations during medical consultation recaps.
- Automated structured Electronic Medical Record (EMR) auto-filing and diagnostic entity extraction via a responsive **React.js** clinician dashboard, streamlining post-consultation documentation workflows.

RateMyCareer — **Project AI Implementation Lead** (Python, Flask, React.js, Supabase, RoBERTa) Fall 2025

- Directed technical architecture for a 6-person team in the UTD AI Mentorship program, reducing sentiment misclassification by ∼45% by upgrading the ML inference pipeline from VADER to **RoBERTa** with custom tokenization.
- Eliminated manual moderation overhead by engineering an automated ETL pipeline that extracted, preprocessed, and stored 10,000+ social media posts in **Supabase**, defending the end-to-end architecture before engineers from **Google** and **Apple**.

LEADERSHIP

Team Captain — *FIRST Tech Challenge Team 8626 (Innovate & Design Award Winner)* Spring 2025

- Directed Agile development of a state-championship-qualifying robot while establishing a **Java** and mechanical design mentorship pipeline for 20+ students.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL (PostgreSQL, SQLite), HTML5, CSS3
Frameworks & Libraries: Next.js, React.js, React Native, FastAPI, Node.js, Express, Flask, Tailwind CSS, aiosqlite
AI & Data Science: PyTorch, TensorFlow, scikit-learn, RoBERTa, VADER, OpenCV, NumPy, pandas
Databases, Cloud & Tools: Supabase, Neo4j, MongoDB, Firebase, AWS (EC2), Docker, Git, Linux (Ubuntu), Figma
Languages & Interests: English (Native), Japanese (JLPT N4 In-Progress); Powerlifting (500 lb Deadlift), Digital Photography, FIRST Robotics Referee

HONORS & AWARDS

- **1st Place**, STREAM Foundation DFW Stock Competition (106% ROI — $2.06M 2-month simulated portfolio)
- **2nd Place**, Lockheed Martin CodeQuest Programming Competition (Advanced League)
- **3rd Place (Figma Track)**, HackAI Hackathon (*Comet Navigator*)
- **Chancellor's Honor Roll**, Dallas College
- **National African American Recognition Award**, College Board